Round 9 - New Estimator Architectures

Designing from the complete error map, not patching V29

Bias-free ReLU MLP, n = 1024, depth = 16, fixed He-Gaussian weights

Target: raw MSE 1.64-1.9e-8 at 133-165 u (or better)

Bottom line. I would stop treating the third cumulant as a bank of birth sources. The complete error map says the failure is fresh and local: the missing signal is the all-distinct post-ReLU cubic field plus a fresh quartic graph field. The most promising replacement is therefore a layer operator that constructs only the contractions the next layer needs. A second, safer design keeps two future-query contractions of the cubic tensor, not the tensor or its sources. Both change the cost law from O(number of sources x n^3) to O(few x n^3) per layer.

Recommendation order: DHMP first; FHQR-2 second. A quadratic-chaos Galerkin design is included as a high-risk third architecture, not as the first implementation target.

Architecture	Persistent high-order state	Classical ledger	L5 ledger	Primary risk
1. Dressed Hypertree Moment Propagation (DHMP)	No K3 source bank; C, exact pair planes, marginals, graph responses	~133-163 u/network	~74-90 u/network	Whether dressed trees + full KxC partner close the all-distinct residual to <2-3%
2. Finite-Horizon Query Response, h=2 (FHQR-2)	2 global n x n cubic query-response matrices + backward queries	~148-163 u/network	~82-90 u/network	Tail closure after horizon 2; query closure must beat the source-bank error
3. Quadratic-Chaos Galerkin (QCG)	First-chaos map + moving second-chaos response matrices	~105-150 u/network	~60-90 u/network	Quadratic basis leakage may recreate an n^3 state

The cost numbers are architecture ledgers, not grader measurements. They use 1 u per conventional dense n^3 product and 0.51 u for a level-5 Strassen product, plus the measured ~13 u/network for covariance + nonlinear closure. All designs must be re-metered in FlopScope before promotion.

1. What the error map forces an architecture to do

- The mean formula is not the problem. A new design should spend essentially no research budget on a new marginal ReLU mean closure.
- The all-distinct post-ReLU K3 field must be created at the ReLU where it is born. A representation that first collapses it to repeated-index stars and hopes future linear mixing recreates it is structurally misaligned with the measured error.
- K4 cannot be a scalar regeneration law. The missing pattern is a fresh connected four-vertex field, especially (2,1,1)/(1,1,1,1), but it is only needed after contraction with the next weight row.
- The coefficient paradox is a warning against adding one Edgeworth family in isolation. Any new architecture should carry a coupled set of diagrams or a renormalized object in which partner terms are generated automatically.
- The new cost must be independent of the number of birth layers. The 116 u young tier and 107 u old tier cannot both survive.

A useful change of viewpoint

For a symmetric cubic tensor K and a query matrix Q, define the two-leg contracted response

$$H_Q(K)[i,p] = \sum_{j,k} K[i,j,k]\, Q[j,p]\, Q[k,p].$$

If Q is the next weight matrix W, the next pre-activation (2,1) slice is obtained without reconstructing K:

$$D21_{next}[p,q] = \sum_i H_{W(K)}[i,p]\, W[i,q] = (H_{W(K)}{}^{T} W)[p,q].$$

This identity is the key architectural escape hatch. V29 transports factor legs and then forms Hadamard products. The proposed architectures contract two tensor legs against the actual future queries before paying for transport.

2. Architecture 1 - Dressed Hypertree Moment Propagation (DHMP)

Status: strongest recommendation. Source-free, graph-based, and directly aligned with the measured fresh-error mechanism.

2.1 State

Object	Shape	Purpose
mu	n	Marginal pre/post mean
C or E	n x n	Exact bivariate covariance edge; E denotes post-ReLU covariance
T = K3(i,i,j)	n x n	Exact repeated-index cubic plane from the bivariate closure
d3	n	Cubic diagonal
K22, K31	n x n	Repeated-index K4 slices already available to the closure
d4	n	K4 diagonal
No source bank	0	No A_b/P_b/T_b objects, no age tiers

Memory is O(n^2). At n=1024, even six float32 dense matrices are only about 24 MiB before work buffers, far below 8 GB.

2.2 The dressed cubic field

For the post-ReLU layer let E be the exact bivariate covariance returned by the bivariate closure. Let a_i = w(1,1)_i and b_i = w(2,1)_i. The weak-correlation Gaussian tree has the familiar center-leaf form b_i a_j a_k C_ij C_ik + permutations. Instead of using the bare edge a_i a_j C_ij, DHMP uses the exact bivariate edge E_ij and preserves the same weak-correlation limit with

$$\tau_i = b_i / \max(a_i^2, \varepsilon).$$

The all-distinct cubic skeleton is

$$K3_{tree}(i,j,k) = \tau_i E_{ij} E_{ik} + \tau_j E_{ji} E_{jk} + \tau_k E_{ki} E_{kj}, \quad \text{for distinct } i,j,k.$$

This is not a low-rank tensor approximation. It is a connected-tree graphical closure with n^2 edge degrees of freedom. Higher Hermite ladders in each pair are already resummed inside E.

For Q = W_next, define

$$R = E Q$$
$$X = (\mathrm{diag}(\tau)\, Q) \circ R$$
$$H_{tree} = \mathrm{diag}(\tau)\,(R \circ R) + 2 E X$$
$$D21_{tree,next} = H_{tree}^T Q.$$

Three dense products produce the all-distinct tree contraction for all n next-layer neurons at once: E Q, E X, and H_tree^T Q.

2.3 Make the repeated planes exact instead of asking the tree model to fit them

The teacher map says the missing information is the strictly-distinct bulk, while the bivariate closure can provide the repeated planes. Therefore DHMP uses the tree model only for the all-distinct sector and adds an exact repeated-plane correction.

Compute the tree model's T_tree(i,j)=K3_tree(i,i,j) and d3_tree. Let DeltaT = T_exact - T_tree, with diagonal set to (d3_exact-d3_tree)/3. The symmetric tensor supported only on repeated patterns contracts through Q as

$$U = \Delta T\, Q$$

Repeat_next = (Q elementwise^2)^T U + 2 (Q elementwise* U)^T Q.

This is three more dense products and exactly restores the known iij/iii sector. The architecture therefore spends approximation error only on the sector your teacher says is missing.

2.4 Add the non-Gaussian K3 x C families as a coupled operator, not a coefficient patch

The two measured first-order K3 x C families can be generated from the ARC vector-partition table and compiled directly into contractions of T, C/E and Q. Their operands are matrices, so each family can be grouped into one dense product after elementwise Wick weighting. Budget two products per layer for the full pair.

The important architectural rule is that the K3 x C pair is evaluated together with the dressed-edge tree field below. Do not scan its coefficient in isolation in production.

2.5 K4 diagonal from the same graph messages

The ARC factor-K4 implementation organizes the fresh all-distinct fourth-order field into connected path, star and 3+2 hypertrees. That is exactly the structure needed here: contract the graph with Q four times; never build K4(i,j,k,l).

With c_i = w(3,1)_i and sigma_i = c_i/max(a_i^3,eps), reuse R=EQ. The dressed Gaussian star is

k4_star[p] = 4 sum_i sigma_i Q[i,p] R[i,p]^3. (no dense product)

For the path graphs, the two degree-2 vertices carry tau. Set

P = (diag(tau) Q) elementwise* R, Y = E P,

k4_path[p] = 12 sum_i P[i,p] Y[i,p].

This adds one dense product E P for all outputs. The K3 x C 3+2 graph uses the already available cubic two-leg response H and R; its final contraction is a columnwise dot product after the appropriate Wick ratio is applied at the shared vertex. The coefficient is 12 in the ARC graph enumeration. Pairwise K4 level terms (K4 diagonal + K22/K31) are retained exactly; the K22 term requires one dense product against Q elementwise^2.

2.6 The item-6 paradox: what the missing partner probably is

The strongest explanation is diagrammatic double counting. The theoretical K3 x C coefficient is a coefficient in a bare expansion whose two-point edge is a_i a_j C_ij. But your actual chain already uses a near-exact bivariate ReLU closure, so the physical edge is dressed:

E = G + DeltaE, G_ij = a_i a_j C_ij.

A dressed cubic tree contains

tau_i E_ij E_ik = tau_i G_ij G_ik + tau_i(G_ij DeltaE_ik + DeltaE_ij G_ik) + tau_i DeltaE_ij DeltaE_ik.

The second line is a deterministic partner correction. If it is omitted while the bare K3 x C coefficient is restored to its formal value, the update can over-shoot. The shipped half-strength coefficient can then be an accidental counterterm. DHMP includes the partner automatically because its edge is E, not G.

2.7 FLOP ledger

Per transition item	Dense products
Dressed all-distinct cubic tree: R, E X, H^T Q	3
Exact repeated-plane correction: U, (Q^2)^T U, (Q*U)^T Q	3
Full paired K3 x C families	<=2
K4 pairwise level term (K22 vs Q^2)	1
K4 dressed path (E P)	1
K4 star + 3+2 final reductions	0
Total	8-10

Over 15 transitions this is 120-150 u with conventional GEMMs. Adding the measured ~13 u/network covariance + closure ledger gives 133-163 u. If all eligible products use the stated level-5 price, the same arithmetic is about 74-90 u

including covariance + closure. There is therefore enough cost headroom to be wrong by one or two extra GEMMs per layer and still live in the leader cost band.

2.8 Falsifier - run this before writing an estimator

1. Teacher-force one layer at a time. Use true current post E, T, d3, K22, K31, d4 and the actual next W. Build DHMP D21_next and d4_next. Compare to the stored true next pre-activation slices.
2. Cubic kill gate: relative RMS(D21_DHMP - D21_true) / RMS(D21_true) must be <= 2.5% at layers 4-14. Also require that the error is not strongly depth-growing. If it is >4% at two mid/deep layers, kill DHMP.
3. Paradox test: compare bare-edge and dressed-edge versions while sweeping only the K3 x C coefficient. Promotion signal: with dressed edges, the optimum moves toward the formal full coefficient and the fresh D21 error drops. If the optimum remains near the shipped 1.25x and dressing is neutral, the proposed partner explanation is wrong.
4. Quartic gate: on |mu/sigma|<1.5 neurons, require K4-diagonal correlation >=0.90 and relative RMS <=12% using teacher-forced inputs. The current ~20% error leaves enough room that this is a meaningful gate.
5. Only after those one-step gates pass, run end-to-end on the two teacher networks, then the 100-network harness. Promotion target: raw <=1.9e-8 at <=165 u. Stretch target: <=1.8e-8 at <=120 u.

3. Architecture 2 - Finite-Horizon Query Response (FHQR-2)

Status: second recommendation. More state than DHMP, but it preserves young cubic memory in the exact query directions that will actually be used.

3.1 Why this is different from the adjoint/source approaches already killed

A per-source adjoint still pays one contraction for every source-layer pair. FHQR performs the sum over sources first. It stores global contractions of the entire cubic tensor against known future queries. The state count depends on a short horizon h, not on the number of births.

3.2 State and backward query pass

After a cheap forward pass supplies the derivative diagonals a_l=w(1,1), construct tangent queries backward. For h=2, keep

$$Q_l^{(1)} = W_{l+1}$$

$$Q_l^{(2)} = W_{l+1}\,\mathrm{diag}(a_{l+1})\,W_{l+2}$$ (orientation adjusted to the implementation convention).

The forward high-order state is two n x n response matrices

$$H_l^{(t)}[i,p] = \sum_{j,k} K3_post_l[i,j,k]\,Q_l^{(t)}[j,p]\,Q_l^{(t)}[k,p], \quad t=1,2.$$

For h=2 this is two global matrices, regardless of how many birth layers contributed to K3.

3.3 Propagation rule

6. Linear shift: because Q_l^(2) already contains the next tangent map, the linearly transported old cubic content needed one layer later is obtained from H_l^(2) by a first-leg multiplication with W_{l+1}; no A_b/P_b legs are transported.
7. Nonlinearity: multiply the old response by the appropriate ReLU first-derivative factors and add the fresh dressed-tree birth directly in each query Q^(t). The fresh term uses the same two products E Q and E((tau Q)*R) as DHMP.
8. Readout: D21_{l+1} is H_l^(1)^T W_{l+1} plus the exact repeated-plane correction. The K4 3+2 term reuses H_l^(1); the Gaussian K4 path/star use the same E Q message.
9. Horizon tail: information older than h steps is not dropped. It is closed by the DHMP dressed tree operator. Thus h controls how much non-Gaussian memory is exact in query space, while the tail has constant cost.

3.4 Why this specifically attacks the 116 u young tier

Young source legs are expensive because V29 carries two dense n x n legs for every recent birth even though the next layer consumes only quadratic queries of the tensor. FHQR stores those quadratic queries directly. It therefore preserves the part of a young source that is visible to the next h layers without carrying the source identity or suffering the Khatri-Rao reformation wall.

3.5 FLOP ledger for h=2

Item	Dense products per transition
Backward construction of the second query	~1
First-leg transport of H^(1), H^(2)	~2
Fresh dressed-tree birth for two query families	~4
D21 readout H^(1)^T W	1
K4 path/level sidecar	~1-2
Total	~9-10

This gives about 148-163 u/network including the measured covariance + closure ledger with conventional products, or about 82-90 u with the stated level-5 product price. State memory is small: two H matrices + one extra Q matrix + the ordinary pairwise state is tens of MiB, not GiB.

3.6 Falsifier

10. Use stored V29 source factors only as an oracle, not as the proposed state. At layers 3, 7, 11 and 14, contract the full source sum directly into H_Q for Q^(1) and Q^(2). This is a one-time numpy teacher calculation.
11. Verify the query-shift identity numerically through the next linear layer. This should be machine-precision if indexing and derivative placement are correct.
12. Then replace the exact horizon tail by the dressed-tree tail. Kill h=2 if the resulting next D21 error exceeds 2.5% RMS or if h=2 improves pure DHMP by <20% of its remaining error. In that case h=3 is unlikely to justify its extra products unless the error decay from h=1 to h=2 is close to geometric.
13. Optional independent test: the public width-256 full-K3 tensors can be contracted into H_Q directly. This avoids trusting V29 factors when validating the query state.
14. End-to-end ladder: h=0 (DHMP), h=1, h=2, h=3 only on layers 8-15. Promote the cheapest point with raw <=1.9e-8 and score improvement on the paired 100-network harness.

4. Architecture 3 - Quadratic-Chaos Galerkin (QCG)

Status: high-risk research architecture. It is attractive because K3 and K4 are generated by one common stochastic state, so inconsistent coefficient patches disappear. I would run only its falsifier before implementation work.

4.1 Idea

Write each centered neuron as a truncated Wiener-Hermite field rather than as cumulants born at layers:

$$h_i - E[h_i] \approx u_i^T \xi + (\xi^T H_i \xi - \mathrm{tr}(H_i)).$$

A quadratic Gaussian field automatically produces covariance, third cumulants and fourth cumulants through contractions of the same u_i and H_i. The architecture therefore cannot independently mis-tune a K3 x C family without also changing the partner quartic terms; this is exactly the consistency missing in item 6.

4.2 Compressed Galerkin state

The impossible version stores one full H_i per neuron (O(n^3) state). The testable version uses a moving common quadratic basis B with n modes and stores only coefficient matrix S:

$$H_i \approx \sum_r S[i,r]\, b_r b_r^T, \qquad B = [b_1 \dots b_n].$$

State: U (n x n first-chaos map), B (n x n quadratic basis), S (n x n coefficients), plus mean. This is O(n^2) memory. A linear layer mixes rows of U and S. A ReLU adds a local rank-one quadratic birth proportional to w(2,1)_i u_i u_i^T, then the quadratic field is projected back into the moving B basis.

4.3 Cost and the dangerous step

A practical implementation would spend roughly 6-9 dense products per transition: first-chaos transport, coefficient mixing, one overlap/rebasis product, and 3-6 Gram contractions to obtain covariance/K3/K4. That is roughly 105-150 u/network conventionally including closure, or 60-90 u at the stated L5 price.

The dangerous step is the re-basis projection. A fresh square (u_i^T xi)^2 creates cross terms in a generic basis. If the off-diagonal quadratic-core energy is not small, the required basis dimension grows to n(n+1)/2 and the architecture collapses back to an impossible state. This is why QCG is third, not first.

4.4 Falsifier

15. On a width-256 public full-K3 tensor, fit the best n-mode common quadratic-basis model at layers 0, 4, 8 and 16 using only an evaluation split. Measure tensor contraction error after the next W, not raw Frobenius error.
16. Kill if the best n-mode quadratic basis cannot reach <=3% contracted D21 error at the early layers. Public evidence already says early K3 is much less compressible than deep K3, so this is a severe gate.
17. If it passes, implement the one-network forward Galerkin chain at n=1024 and compare var, d3 and d4 teacher channels. Require simultaneous improvement in all three; improvement in K3 with degradation in K4 is a direct failure of the motivation.

5. Which architecture I would build first

Rank	Architecture	Reason	Stop condition
1	DHMP	It attacks the measured fresh all-distinct K3 and fresh K4 exactly where they are born, has no source count, and the one-step falsifier uses arrays you already have.	Teacher-forced D21 error >4% at two mid/deep layers, or dressed-edge paradox test fails.
2	FHQR-2	If DHMP is close but not enough, this preserves two layers of cubic memory in the exact future directions while still avoiding source identities.	h=2 captures <20% of DHMP residual or costs >165 u after real metering.
3	QCG	The only proposal here that makes K3 and K4 manifestations of one common stochastic state; theoretically elegant but closure-risky.	Early-layer common quadratic basis needs more than n modes for <=3% contracted error.

Expected score economics

These are promotion targets, not predictions. If DHMP reaches raw 1.8e-8 at 118 u, the score is about 2.07e-9. At <=102.4 u the competition multiplier hits the 0.1 floor, so the same raw error scores 1.8e-9. FHQR-2 at raw 1.8e-8 and 148 u scores about 2.60e-9. Therefore DHMP should spend any sub-floor headroom on accuracy rather than further arithmetic trimming.

The first experiment I would run

Do not implement an estimator yet. Build a pure numpy one-step notebook with teacher-forced layer inputs and the actual next W. It should output four curves across depth:

- bare tree D21 error;
- dressed tree D21 error;
- dressed tree + exact repeated-plane correction;

- dressed tree + repeated correction + full K3 x C pair.

For each curve also compute the next K4 diagonal using star/path/3+2 contractions. This single notebook decides whether the source-free architecture is real. If the final curve is <=2.5% D21 RMS and <=12% active-neuron K4 RMS at the mid/deep teacher layers, proceed directly to DHMP. If not, run the FHQR h=1/h=2 oracle contraction before abandoning the architecture family.

6. Minimal DHMP pseudocode

```
for l in 0..L-2:
    # existing exact/near-exact bivariate state
    mu_pre, C_pre, T_pre, d3_pre, K22_pre, K31_pre, d4_pre = state
    post = exact_bivariate_relu_closure(state)
    E, T, d3, K22, K31, d4 = post.pair_state
    Q = W[l+1]

    # dressed all-distinct cubic tree
    tau = w2 / maximum(w1*w1, eps)
    R = E @ Q
    X = (tau[:,None] * Q) * R
    H = tau[:,None] * (R*R) + 2 * (E @ X)
    D21 = H.T @ Q

    # exact repeated-plane replacement
    T_model, d3_model = pair_planes_of_tree(E, tau)
    Delta = T - T_model
    diag(Delta) = (d3 - d3_model)/3
    U = Delta @ Q
    D21 += (Q*Q).T @ U + 2 * (Q*U).T @ Q

    # add the two full K3 x C vector-partition families
    D21 += compiled_KC_pair(T, E, Q, wick_coeffs)

    # quartic diagonal: exact repeated level + dressed fresh graphs
    q2 = Q*Q
    k4 = exact_pairwise_k4_level(d4, K22, K31, Q)
    sigma = w3 / maximum(w1*w1*w1, eps)
    k4 += 4 * colsum((sigma[:,None]*Q) * (R**3))
    P = X
    Y = E @ P
    k4 += 12 * colsum(P*Y)
    k4 += contracted_3plus2_from_H(H, R, Q, wick_coeffs)

    state_next = build_next_pre_state(mu_post, C_post, D21, k4, Q, ...)
```

7. What would falsify the entire thesis

- If teacher-forced current pair statistics do not predict the next preactivation D21 through any dressed tree/KxC graph operator to within ~3%, then the missing all-distinct field has genuine long memory that cannot be summarized by pair state. DHMP is then dead.

- If FHQR h=2 does not materially reduce that residual, the useful memory is not short-horizon in query space. At that point the source-bank floor is probably a real information-complexity floor for deterministic analytic methods at this accuracy.
- If the paradox persists after replacing bare edges by exact bivariate dressed edges, then the missing partner is not pair-edge dressing. The next suspects are higher connected K4/K5 blocks rather than a coefficient issue.
- If K4 path/star/3+2 predicts the teacher pattern but end-to-end MSE worsens, the chain contains trajectory-level error cancellation. Then promotion must be based only on 100-network paired score, not local channel accuracy.

8. External anchors and why they matter

These sources do not publish the architectures above. They support the algebraic ingredients or provide negative evidence that constrains the designs:

[1] Alignment Research Center, "Cumulant propagation for wide random MLPs" (public code). https://github.com/alignment-research-center/mlp_cumulant_propagation - The factor_k4 implementation explicitly organizes connected fourth-order content into path, star and 3+2 hypertree diagrams, which motivates direct graph contraction.

[2] G. Peccati and M. S. Taqqu, Wiener Chaos: Moments, Cumulants and Diagrams, Springer, 2011. https://doi.org/10.1007/978-88-470-1679-8 - General diagram/cumulant calculus for nonlinear functionals of Gaussian fields.

[3] F. Voigtlaender, "A General Version of Price's Theorem," Journal of Theoretical Probability 34 (2021). https://doi.org/10.1007/s10959-020-01017-w - Justifies differentiating Gaussian nonlinear expectations with respect to covariance; conceptually supports dressed bivariate response edges.

[4] K. Fischer et al., "Decomposing neural networks as mappings of correlation functions," Physical Review Research 4, 043143 (2022). https://doi.org/10.1103/PhysRevResearch.4.043143 - Uses Price-theorem/correlation-function expansions for nonlinear networks; useful context for graph expansions, though it does not solve this high-accuracy fixed-weight problem.

[5] jamesrahenry/arc-whitebox-replication, public full-K3 evaluation. https://github.com/jamesrahenry/arc-whitebox-replication - Independent evidence that strictly-distinct K3 carries most recoverable signal and that a naive local/windowed analytic carrier is not accurate enough; this is why DHMP uses exact pair-plane replacement and FHQR uses future queries instead of a bare local source.

[6] J. Exenberger, S. Ranftl, R. Peharz, "Deep Polynomial Chaos Expansion," AISTATS 2026. https://proceedings.mlr.press/v300/exenberger26a.html - Motivation for the high-risk Galerkin alternative: structured polynomial-chaos representations can avoid explicit high-dimensional basis enumeration, but applicability here must be falsified early.

9. Final recommendation

Build the DHMP one-step falsifier first. It is the only proposal that simultaneously (i) removes the source-count cost, (ii) creates the all-distinct cubic field at the ReLU where the teacher says the error is born, (iii) obtains the K4 star/path/3+2 correction from the same messages, and (iv) gives a concrete explanation for the coefficient paradox. If DHMP is close but misses the 2-3% cubic target, add FHQR-2 rather than restoring source identities. I would not spend implementation time on QCG unless its basis-leakage falsifier passes.

Important: the architectures are new proposals derived from the supplied error map and public diagram identities. They are not claimed as published leaderboard methods. The raw-MSE targets in the report are promotion thresholds, not predictions.